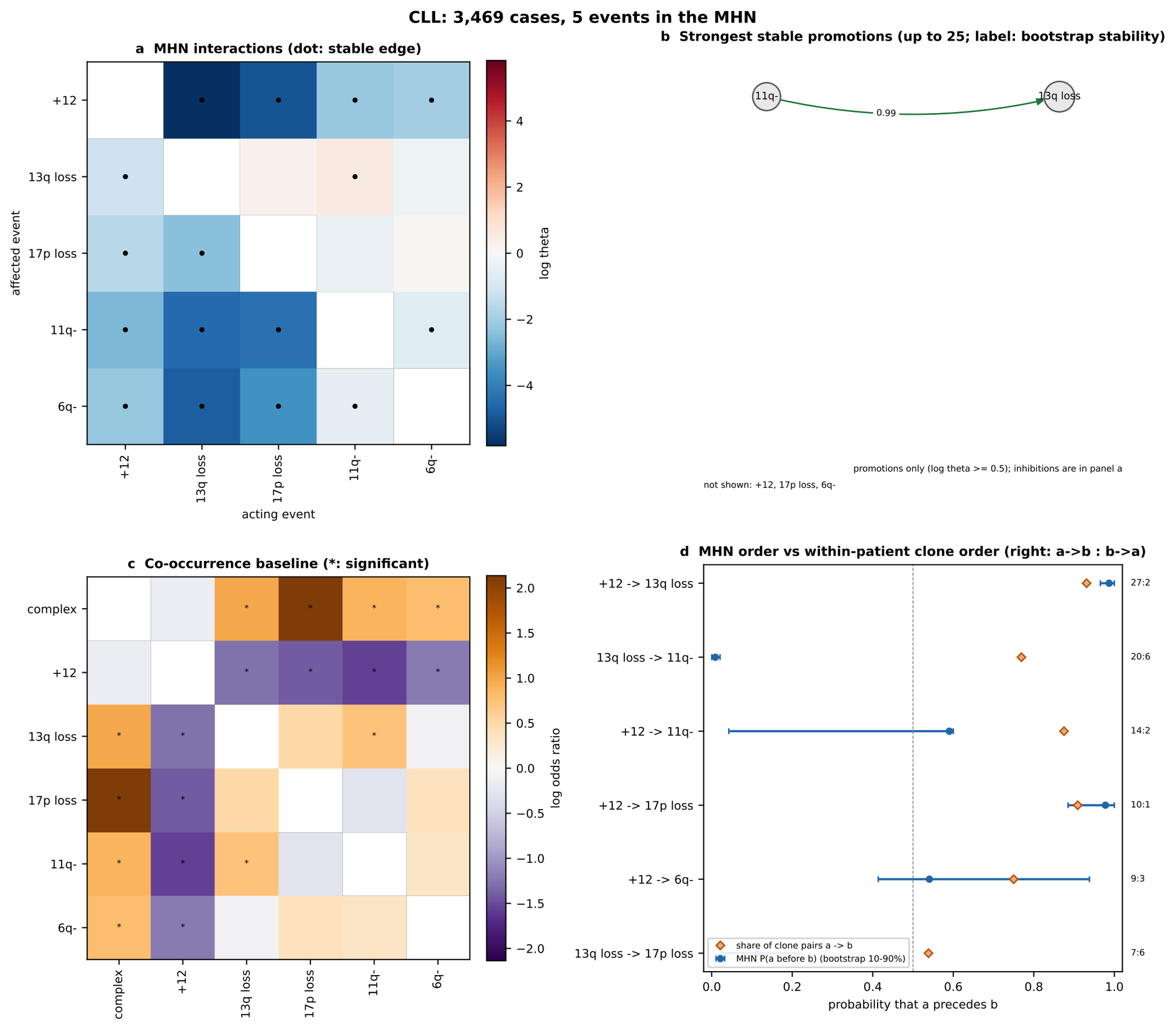

CLL: 3,469 cases, 5 events in the MHN
a MHN interactions (dot: stable edge)
affected event
+12
13q loss
17p loss
11q-
6q-
acting event
log theta
4
2
0
−2
−4
b Strongest stable promotions (up to 25; label: bootstrap stability)
11q-
13q loss
0.99
promotions only (log theta >= 0.5); inhibitions are in panel a
not shown: +12, 17p loss, 6q-
c Co-occurrence baseline (*: significant)
complex
+12
13q loss
17p loss
11q-
6q-
log odds ratio
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
d MHN order vs within-patient clone order (right: a->b : b->a)
+12 -> 13q loss
13q loss -> 11q-
+12 -> 11q-
+12 -> 17p loss
+12 -> 6q-
13q loss -> 17p loss
27:2
20:6
14:2
10:1
9:3
7:6
share of clone pairs a -> b
MHN P(a before b) (bootstrap 10-90%)
0.0
0.2
0.4
0.6
0.8
1.0
probability that a precedes b